Question 1:

1. (24 points total) For this question you are going to provide code fragments that implement two methods. You do not need to provide anything other than those two methods.

   Imagine using a doubly linked list to implement the stack ADT for values of a generic type. Skeleton code is provided below. Implement the push and pop methods. Instead of using the methods of the List ADT, manipulate the nodes directly. Make sure to handle any major error conditions by throwing a new RuntimeException. This doubly linked list uses sentinel nodes and you can assume that head and tail have already been initialized properly.

```
class LinkedListStack<E> {

    private static class Node<E> {
        public Node(E d, Node p, Node n) {
            data = d; prev = p; next = n;
        }

        Node prev;
        Node next;
        E data;
    }

    Node head;
    Node tail;

    public void push(E x) { /* write the code for this in the file that you will upload */ }
    public E pop() { /* write the code for this in the file that you will upload */ }
}
```

```
public push(E x) {
        Node<E> n = new Node<>(x, tail.prev, tail);
        tail.prev.next = n;
        tail.prev = n;

        Node<E> n = new Node<>(x, head, head.next);
        head.next.prev = n;
        head.next = n;
}

public E pop() {
        if (head.next == tail) {
                // can either return null or throw an exception
                return null; OR throw new EmptyStackException();
        }
        Node temp = tail.prev;
        temp.prev.next = tail;
        tail.prev = temp.prev;
        return temp.data;
```

```
        Node temp = head.next;
        temp.next.prev = head;
        head.next = temp.next;
        return temp.data;
}
```

Question 2:

2. (12 points total) You are given both the pre-order traversal and an in-order traversal for a unique binary tree.

   Pre-order traversal: A C E B F D

   In-order traversal: E C F B A D

   - Draw the unique tree defined by those traversals.
   - Write down the corresponding post-order traversal for that tree.

Unique tree:

```
          A
       /.    \
      C         D
    /   \
   E.     B
        /.
        F
```

Postorder:
E F B C D A

Question 3:

3. (14 points total) Recall the firstChild/nextSibling representation of general tree. A node in the tree is represented by the following node object:

```
class TreeNode<E> {

    E data;
    TreeNode<E> firstChild;
    TreeNode<E> nextSibling;
}
```

Fill in the following method. The method's job is to print out the tree by performing a **pre-order** traversal. Each node's data can be printed out on its own line, there is no need to indent any of the nodes. You can assume that the data has a useful toString method.

```
public void preorder(TreeNode<E> root)  { /* write the code here */ }
```

```
public void preorder(TreeNode<E> root)
{

    if(root==null)
        return;
    System.out.println(root.data);
    preorder(root.firstChild);
    preorder(root.nextSibling);

}
```